Question ID: 7f87deff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

What is an *x*-coordinate of an *x*-intercept of the graph of $y = 3(x-14)(x+5)(x+4)$ in the *xy*-plane?

Correct Answer: 14, -5, -4

Rationale

The correct answer is either 14, -5, or -4. The *x*-intercepts of a graph in the *xy*-plane are the points at which the graph intersects the *x*-axis, or when the value of y is 0. Substituting 0 for y in the given equation yields $0 = 3(x-14)(x+5)(x+4)$. Dividing both sides of this equation by 3 yields $0 = (x-14)(x+5)(x+4)$. Applying the zero product property to this equation yields three equations: $x - 14 = 0$, $x + 5 = 0$, and $x + 4 = 0$. Adding 14 to both sides of the equation $x - 14 = 0$ yields $x = 14$, subtracting 5 from both sides of the equation $x + 5 = 0$ yields $x = -5$, and subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the *x*-coordinates of the *x*-intercepts of the graph of the given equation are 14, -5, and -4. Note that 14, -5, and -4 are examples of ways to enter a correct answer.